Problem Set 4
ECO3121 - Fall 2024

Due 3 PM, December 11, 2024
No late submission is allowed
Please combine your answer, Stata code and requested output in **one** pdf file and upload it to Blackboard

Question 1

A researcher is interested in analyzing the effect of a free fertilizer policy on crop yield. He has a panel data set for 1000 villages in rural China over 2016 to 2019 in which the data for the average crop yield (Y_{it}) in village (entity) i in (time) year t and the indicator for whether the village participate in this free fertilizer program (X_{it}) are available.

1. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + u_{it}$$

   where α_i are i.i.d. unobserved random variables and α_i is correlated with X_{it}. (a) Explain how the researcher should estimate β_1 with the data he has. (b) Provide a factor that is modelled by α_i. (2 points)

2. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \gamma_t + u_{it}$$

   where γ_t are i.i.d. unobserved random variables and γ_t is correlated with X_{it}. (a) Explain how the researcher should estimate β_1 with the data he has. (b) Provide a factor that is modelled by γ_t. (2 points)

3. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \gamma_t + u_{it}$$

   What is the advantage of this model compared to the above two? (2 points)

4. The fixed effects estimator of β_1 in (3) can be obtained by applying two-way demean on this model. The first is entity-demean ignoring the time fixed-effects followed by the time-demean ignoring the entity-fixed effects. Assume the panel is balanced.

   Show that the order of demean is unimportant (either starting with entity-demean or time-demean doesn't matter). Give an intuitive explanation for why. (3 points)

Question 2

Following a national poverty alleviation program in Gansu province, in July 2017, many households in Gansu province received subsidy and financial support. Thus, this policy experiment induced a geographical allocation of subsidy that can be presumed exogenous in an income growth. Let Y_i^0 and Y_i^1 denote the consumption in village i before and after the policy intervention. Let D_i be a dummy variable that takes the value 1 if household i is in Gansu province, 0 otherwise (other provinces). We would like to estimate the treatment effect of the income change on consumption with the difference-in-difference estimator.

1. (2 points) Write down this difference-in-difference estimator as a function of $\{Y_i^0, Y_i^1, D_i\}$ for $i = 1, ..., n$.

2. (2 points) What is the key assumption for this estimator to be an unbiased estimator of the treatment effect?

3. (2 points) Suppose after the policy intervention, households in Gansu province works harder (policy irrelevant) and thus higher income and consumption are expected. Does the difference-in-difference estimator under or over estimate the treatment effect. Explain.

Question 3

We are going to replicate a study conducted by Card and Krueger in 1994 that investigate the relationship between a rise in minimum wage and employment.

Economic theories have long suggested that increases in the minimum wage lead to a reduction in the employment for at least two reasons: Businesses are less likely to hire and will rather invest in other resources that are now cheaper because of wage increase. Higher salaries will induce businesses to raise their prices to compensate their greater costs; as prices increase, we expect fewer buyers, which will lead to lower demand and employment.

These theories have found mixed support but the discussion is still very much open within the policy world, as states discuss the opportunity to rise their minimum wage to help local populations to face increasing living costs. Discussions are currently occuring in New Jersey and Illinois to raise the minimum wage to 15$/hour (New york has successfully passed this same raise in 2018).

One of the first study looking at this policy problem was Card and Krueger's. They applied a difference-in-difference the design to look at two groups of fast-food restaurants: fast-food restaurants in New Jersey where the minimum wage increased from 4.25$ to 5.05$ per hour (treatment group) AND fast-food restaurants in Pennsylvania where the minimum wage did not increase (control group). They collected employment data before and after the minimum wage was approved.

Table 1: Variable Description

Variable name	Description
ID	Unique identifier for fast food
Treatment	Pre-treatment (=0) and post-treatment (=1)
Group	1 if NJ (treatment); 0 if PA (Control)
Empl	# of full time employees
C.Owned	If owned by a company (=1) or not (=0)
Hours.Opening	Number hours open per day
Soda	Price of medium soda, including tax
Fries	price of small fries, including tax
Chain	1 = BK, 2 = KFC, 3 = Roys, 4 = Wendys
SouthJ	South New Jersey
CentralJ	Central New Jersey
NorthJ	North New Jersey
PA1	Northeast suburbs of Philadelphia
PA2	Easton and other PA areas
Shore	New Jersey Shore

1. Explain why a simple comparison in employment between New Jersey and PA after the minimum wage policy may not be a good estimation for the treatment effect. (2 points)

2. What is the regression model you'd like to estimate? (2 points)

3. What is the key assumption of the estimation method you use? (1 point)

4. The data is in the blackboard (bb) Assignment 4 file folder titled *DID_Example.csv*, and the variables are defined as in Table 1. Run the regression model you proposed in 2, and report the regression results. (3 points)
   (Hint: You need first import the csv file into Stata.
   Replace all "NA" with the missing value in variables, using

   replace var_name = "" if var_name == "NA"

   Then, convert string variables to numeric variables, by the "*destring*" command. Please include control variables that you think necessary.)

5. Explain why this is an unbiased estimation of the treatment effect. (Hint: You can use a graph if necessary) (2 points)